Assignment 1: Pentium M Hybrid Branch Predictor

Bimodal, global and loop predictors on the CBP-2 trace infrastructure

1. Summary

I implemented the branch-direction part of the Pentium M predictor in the `pm_predictor` class in `cbp2-infrastructure-v2/src/my_predictor.h`. It has three components:

- a **4096-entry bimodal table**;
- a **tagged, 4-way, 512-set global predictor** (required by the assignment);
- a **128-entry loop predictor** (optional in the assignment; added as an extension).

I evaluated it on all 20 CBP-2 traces.

Predictor	Storage	Average MPKI (20 traces)
Always taken	0	86.312
Bimodal table alone (4K x 2-bit)	1 KB	10.267
gshare, 32K entries (the sample code in the kit)	8 KB	6.306
Pentium M bimodal + global (required part)	~3.75 KB	5.849
Pentium M bimodal + global + loop (submitted)	**~4.1 KB**	**5.649**

The submitted predictor has **45% fewer mispredictions than the bimodal table alone**. It is **10% better than the sample 32K-entry gshare while using about half its storage**. The loop predictor removes another 3.4% of mispredictions compared with bimodal + global, and improves 17 of the 20 traces.

Verification:

- **Worked example.** A unit test replays the 22-branch example in the handout (Appendix B). It matches every column of the results table and two full snapshots of the global table, LRU bits included.
- **Loop predictor.** A second test checks it learns every trip count from 1 to 63 exactly and never uses loops too long for its 6-bit counters. It also checks re-learning after a trip-count change, the index/tag/LRU fields, and that it removes the loop-exit mispredictions end to end.

2. What was implemented

2.1 Structure

The design follows the Pentium M diagram in the handout. The sizes come from Uzelac & Milenkovic, *Experiment Flows and Microbenchmarks for Reverse Engineering of Branch Predictor Structures* (ISPASS 2009).

Component	Organization	Index / tag
Bimodal table	4096 x 2-bit saturating counters	index = IP[11:0]
Global history	15-bit global outcome history register (GHR)	shift in each conditional outcome
Global predictor	512 sets x 4 ways = 2048 entries; entry = {valid, 6-bit tag, 2-bit counter} + LRU	HASH = IP[14:0] XOR GHR; set = HASH[14:6], tag = HASH[5:0]
Loop predictor buffer (LPB)	64 sets x 2 ways = 128 entries; entry = {valid, 6-bit tag, 6-bit Count, 6-bit Limit, 1-bit Prediction, 2-bit confidence} + LRU	set = IP[9:4], tag = IP[15:10]
Final prediction	confident LPB hit → loop prediction; else global hit → global counter; else bimodal counter	

As the handout allows, the path information register (PIR) and its hash are replaced by a plain global outcome history. This is the same gshare-style XOR used in the worked example. A PIR variant is also included (section 4.5). Branch target prediction (BTB/iBTB) is not implemented, as the assignment permits, so `target_prediction` is always 0.

Storage:

- Bimodal: 4096 x 2 b = 1 KB.
- Global: 2048 x (1 + 6 + 2 + 2 LRU) b = 2.75 KB.
- LPB: 128 x (1 + 6 + 6 + 6 + 1 + 2 + 1 LRU) b = 368 B.
- Total: about 4.1 KB.

2.2 Prediction (`predict`)

For a conditional branch:

1. **Bimodal:** read the counter at `IP[11:0]`. The prediction is its upper bit.
2. **Global:** compute `HASH = (IP ^ GHR) & 0x7FFF`, the set `HASH>>6` and the tag `HASH & 0x3F`. Compare the tag against all 4 valid ways of that set.
3. **Loop:** look up set `IP[9:4]` and tag `IP[15:10]`. On a hit, the loop prediction is `Count == Limit ? !Prediction : Prediction`. It is *used* only if the entry is confident: its trip count has repeated 3 times in a row.
4. **Choose:** use the confident loop prediction if there is one. Otherwise use the global prediction on a global hit. Otherwise use the bimodal prediction.

Non-conditional branches are predicted taken and do not touch any table. The simulator only scores conditional branches.

2.3 Update (`update`)

- **Bimodal** – the counter is always trained with the outcome.
- **Global, hit** – the hit entry's counter is trained, and the entry becomes most-recently-used.
- **Global, miss** – a new entry is allocated if the prediction that was available *without* the global table was wrong. That is the loop prediction if one was used, otherwise the bimodal prediction. The victim is an invalid way if any, otherwise the LRU way. The counter starts weakly biased toward the outcome (2 if taken, 1 if not taken).
- **Loop** – see 2.4.
- **GHR** – the outcome is shifted in (15 bits).

LRU is exact in both set-associative tables. Each set keeps a permutation of ages, and a hit or an allocation makes that way the most recently used. All sizes and policies are fields of `pm_config`. The default constructor, which `predict.cc` uses, builds the submitted configuration. Setting `use_loop = 0` gives the bimodal + global predictor.

2.4 Loop predictor

An LPB entry describes a branch that goes in its *Prediction* direction *Limit* times in a row and then once the other way (the loop exit), over and over. *Count* is the number of Prediction-direction outcomes since the last exit.

- **Allocate.** On an LPB miss where the bimodal prediction was wrong, an entry is allocated. This is the paper's "first opposite outcome", for example the first exit of a loop whose body the bimodal table predicts as taken. *Prediction* is set to the direction that was just broken, *Count* = 0, *Limit* is unknown, and the victim is the LRU way.
- **Inside the loop** (outcome = Prediction): `Count++`.
  - If Count goes past Limit, the loop ran longer than learned, so confidence is reset.
  - If Count overflows its 6 bits, the loop is too long to track. Limit becomes unknown and the entry is never used.
- **Exit** (outcome ≠ Prediction):
  - If Count = Limit, the trip count repeated, so confidence is incremented (saturating at 3).
  - Otherwise Limit = Count and confidence is reset.
  - Count returns to 0.
- **Re-orientation.** Two exits in a row (Count = 0) mean the entry is pointing the wrong way, for example because it was allocated while the bimodal counter was still cold. Prediction is flipped and the entry retrains.

2.5 Deliberate differences from the handout and the paper

1. **Global allocation (vs. the handout example).** The handout allocates a global entry on *every* global miss and keeps the victim's old counter. My code supports exactly that (`PM_ALLOC_ALWAYS` + `PM_CTR_KEEP`), and the worked-example test runs in that mode. For the submitted predictor I allocate only when the non-global prediction was wrong. This reserves the global table for branches the cheaper components cannot handle, and cuts MPKI by 25% (section 4.2).
2. **Loop vs. global priority (vs. the paper).** The paper reports that a global hit overrides a loop hit. In this model that makes the loop predictor nearly useless. In a loop longer than the 15-bit history, the history before the exit is all "taken", the same as before every other iteration. The global entry for that history therefore hits at the exit and predicts "taken" again. With global first, the LPB lowers MPKI only from 5.849 to 5.814. With loop first, it reaches 5.649. Both orders are

selectable (`loop_over_global`).

3. **Confidence counter (addition to the paper).** The paper lists only Tag, Count, Limit and Prediction. Without a confidence check, the LPB overrides the bimodal and global predictions for every non-loop branch it captures, giving 8.361 MPKI. I added a 2-bit counter and use the LPB only after the trip count repeats 3 times.
4. **No BTB.** In the real chip an LPB hit also requires a BTB hit. The BTB is not modeled, so this check is treated as always true.

3. Verification

`make test` in `src/` runs two programs (output in `results/test_output.txt`).

`test/test_example.cc` – the handout's worked example. It builds `pm_predictor` at the handout's scale, with the loop predictor off:

- 8-bit addresses and a 16-entry local table indexed by IP[5:2]
- a 4-bit GHR
- 4 rows x 2 ways with 2-bit tags

It uses the handout's initial counter values and its allocation policy. It replays the 22 branch outcomes and checks the GHR, hash, row, tag, local counter, and the local, global and final predictions at every step. It also compares every field of the global table (valid, tag, counter, LRU) with the handout before steps 12 and 22.

```
step br  addr  out  GHR  hash row tag  lctr local global final
   1 b1  0x44  T    0000 0001   0  01     1     N      -     N
   ...
  14 b2  0x6C  NT   1010 0001   0  01     2     T      N     N
   ...
  22 b3  0x90  T    0011 0111   1  11     3     T      -     T
PASS: all 22 steps and both global-table snapshots match the handout.
```

`test/test_loop.cc` – the loop predictor (27 checks):

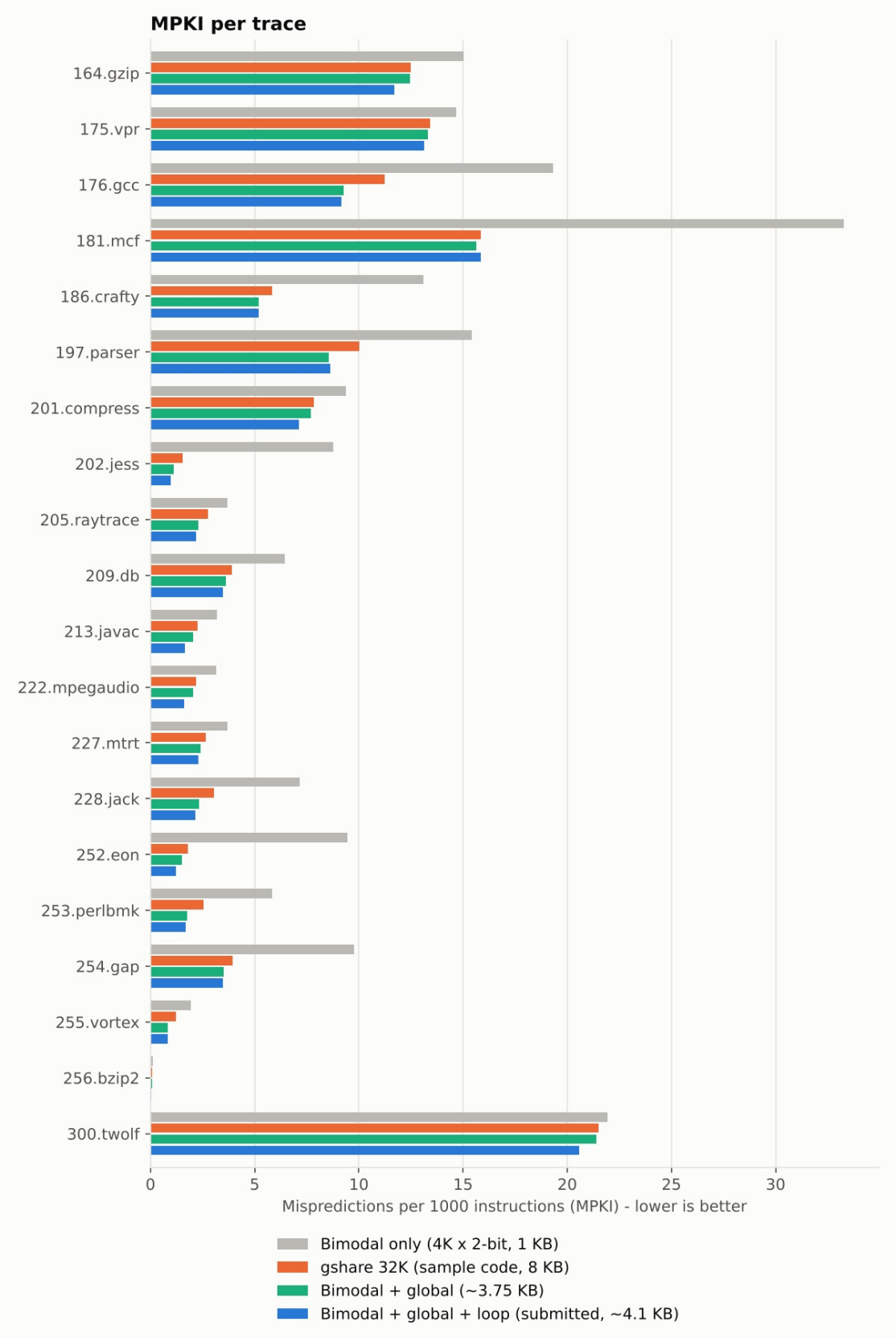
```
taken loop    body= 1 : LPB used   40/  40, LPB misses 0, total misses 0
taken loop    body=10 : LPB used  220/ 220, LPB misses 0, total misses 0
taken loop    body=40 : LPB used  820/ 820, LPB misses 0, total misses 0
taken loop    body=63 : LPB used 1280/1280, LPB misses 0, total misses 0
not-taken loop body=12 : LPB used  260/ 260, LPB misses 0
too-long loops (64, 100): LPB never used
trip count 8 -> 11       : 1 miss, re-learned after 3 confirming runs
LPB index/tag fields, aliasing and LRU replacement OK
end-to-end, 1000 runs of a 30-iteration loop: loop-branch misses 1607 without LPB, 0 with LPB
PASS: all 27 loop predictor checks.
```

I also ran the full predictor through both the kit's `predict` binary (via `run.sh`) and the experiment driver. Both give the same MPKI on every trace.

4. Results: bimodal + global predictor

MPKI is direction mispredictions per 1000 instructions, as reported by `predict` (each trace is 100M instructions). Lower is better.

4.1 Per-trace results


MPKI per trace
164.gzip
175.vpr
176.gcc
181.mcf
186.crafty
197.parser
201.compress
202.jess
205.raytrace
209.db
213.javac
222.mpegaudio
227.mtrt
228.jack
252.eon
253.perlbmk
254.gap
255.vortex
256.bzip2
300.twolf
0
5
10
15
20
25
30
Mispredictions per 1000 instructions (MPKI) - lower is better
Bimodal only (4K x 2-bit, 1 KB)
gshare 32K (sample code, 8 KB)
Bimodal + global (~3.75 KB)
Bimodal + global + loop (submitted, ~4.1 KB)

Trace	Bimodal only	gshare 32K	Bimodal + global	+ loop (submitted)	Change from loop
164.gzip	15.031	12.473	12.439	**11.719**	-5.8%
175.vpr	14.673	13.415	13.298	**13.141**	-1.2%
176.gcc	19.313	11.254	9.266	**9.169**	-1.0%
181.mcf	33.271	15.837	15.624	**15.861**	+1.5%
186.crafty	13.097	5.837	5.208	**5.199**	-0.2%
197.parser	15.420	10.008	8.542	**8.621**	+0.9%
201.compress	9.368	7.831	7.688	**7.117**	-7.4%
202.jess	8.773	1.562	1.124	**0.974**	-13.3%
205.raytrace	3.704	2.756	2.293	**2.201**	-4.0%
209.db	6.437	3.909	3.608	**3.463**	-4.0%
213.javac	3.198	2.267	2.052	**1.649**	-19.6%
222.mpegaudio	3.166	2.188	2.031	**1.617**	-20.4%
227.mtrt	3.682	2.657	2.389	**2.285**	-4.4%
228.jack	7.175	3.033	2.349	**2.151**	-8.4%
252.eon	9.443	1.807	1.506	**1.210**	-19.7%
253.perlbmk	5.848	2.554	1.774	**1.674**	-5.6%
254.gap	9.760	3.926	3.509	**3.484**	-0.7%
255.vortex	1.930	1.222	0.830	**0.835**	+0.6%
256.bzip2	0.113	0.094	0.062	**0.051**	-17.7%
300.twolf	21.943	21.489	21.390	**20.568**	-3.8%
Average	10.267	6.306	5.849	**5.649**	-3.4%

Bimodal + global alone beats the sample gshare on all 20 traces. With the loop predictor added, it does so on 19; mcf is 0.2% behind (15.861 vs. 15.837).

How the two components split the work (bimodal + global, no loop predictor):

- On average the global table supplies 41% of predictions and is right on 92.5% of its hits. Those are mostly the hard, correlated branches.
- The bimodal table supplies the rest and is right 98.1% of the time when used, because the branches it could not handle have been moved to the global table.

Where it helps and where it doesn’t:

- The largest gains over the bimodal table alone are on programs with many correlated branches: gcc (−52%), crafty (−60%), jess (−87%), eon (−84%), perlbmk (−70%) and mcf (−53%).
- twolf, mcf, vpr and gzip stay above 11 MPKI for every configuration tested, with or without the loop predictor. Their branches depend on data and do not correlate well with 15 bits of history.

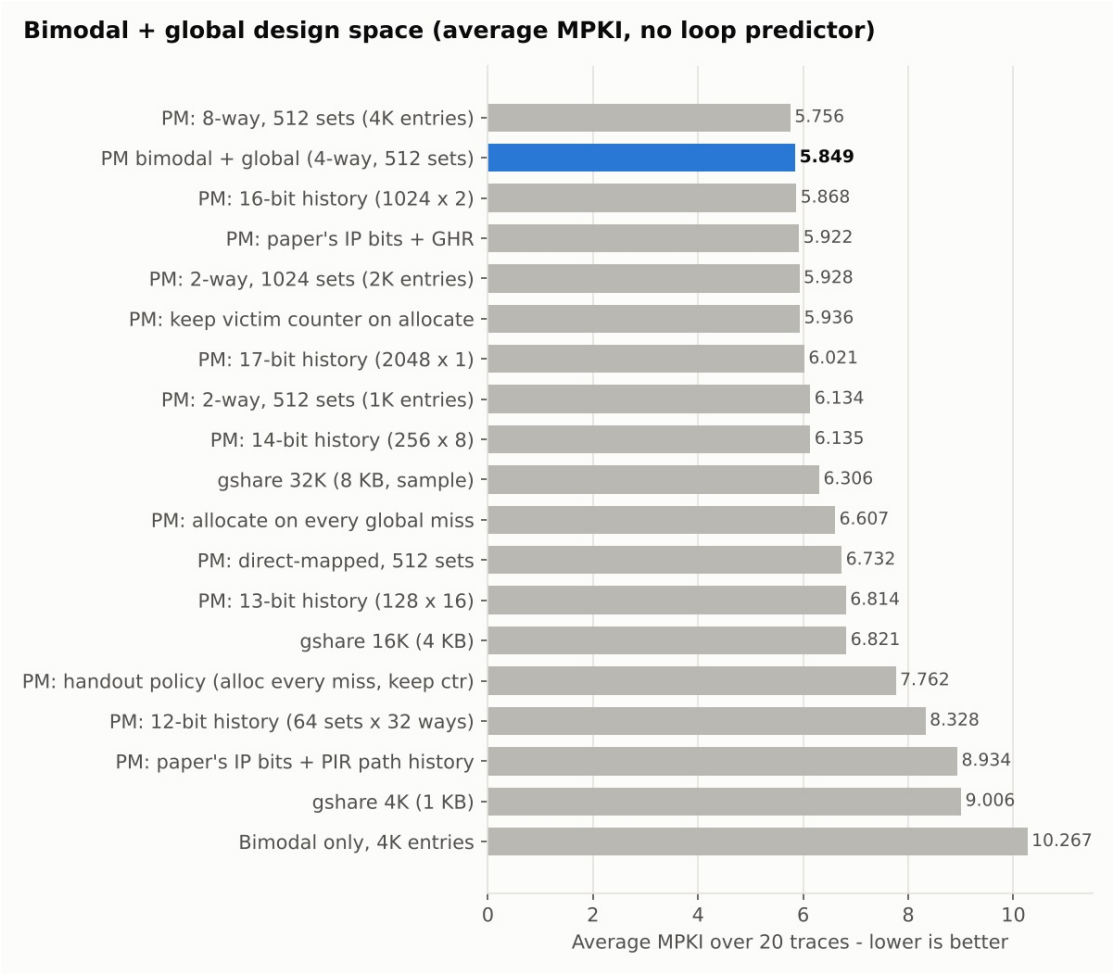
4.2 Allocation policy for the global table

Variant (no loop predictor)	Avg MPKI	Global hit rate	Allocations per trace
Allocate only on bimodal mispredict, weak-init counter	**5.849**	41%	0.15 M
Allocate only on bimodal mispredict, keep victim counter	5.936	41%	0.15 M
Allocate on every global miss, weak-init counter	6.607	87%	1.73 M
Allocate on every global miss, keep victim counter (handout example)	7.762	87%	1.73 M

Allocating on every miss produces almost 12x more allocations. They evict useful entries, and most of the new ones are for easy branches that the bimodal table was already predicting correctly. Filtered allocation spends global capacity only where the

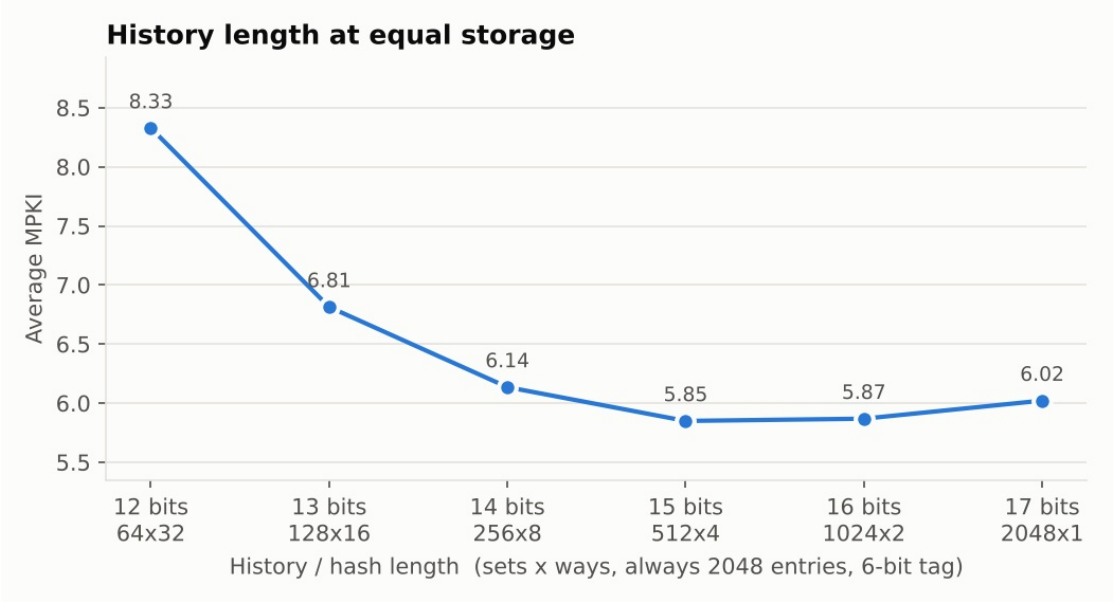
bimodal table fails.

4.3 Associativity, size and hash



- **Associativity.** At 512 sets: direct-mapped 6.732, 2-way 6.134, 4-way 5.849, 8-way 5.756 average MPKI. The 2-way option the assignment allows costs about 5% more mispredictions than 4-way at half the storage. A 2-way table with 1024 sets has the same capacity as 4-way and comes within 1.4% of it (5.928).
- **Hash bits.** I also tried the Pentium M's own IP bit selection: index = IP[12:4]^H[14:6], tag = IP[18:13]^H[5:0]. It scored 5.922, about the same as the plain XOR (5.849).

4.4 History length at equal storage



All points have 2048 entries and a 6-bit tag; only sets x ways changes. Short histories (12–13 bits) do poorly: the hash is only that wide, so few address bits survive and unrelated branches alias. Performance flattens at 15–16 bits. The Pentium M's 15-bit hash with 512 x 4 sits at the minimum.

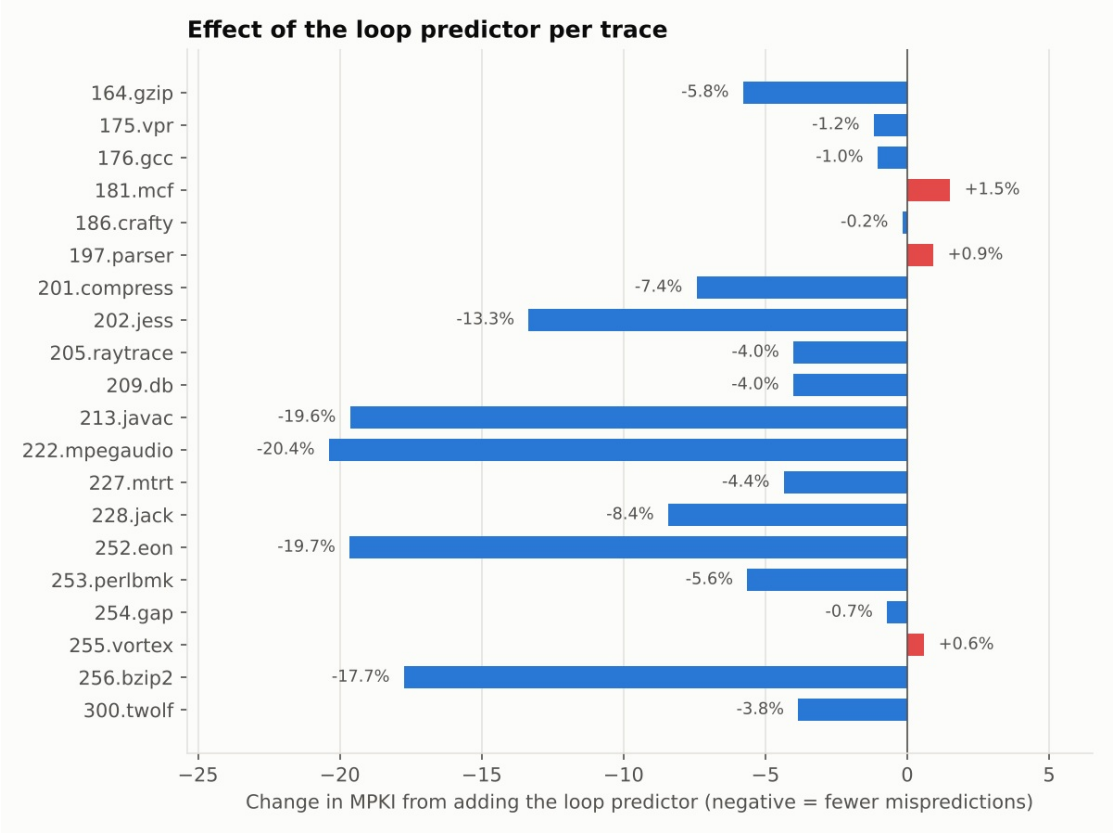
4.5 Real Pentium M path history (PIR)

I implemented the ISPASS 2009 PIR update rule and used it in place of the GHR. On a taken conditional branch, `PIR = (PIR << 2) ^ IP[18:4]`. On an indirect branch, `PIR = (PIR << 2) ^ {IP[18:10], target[5:0]}`.

In isolation it does clearly worse: 8.934 MPKI. It records only about 8 taken branches and no not-taken outcomes. In the real chip it works together with the BTB, which is not modeled. The paper's description of how the hash halves are concatenated is also ambiguous. So I kept the outcome-history GHR, which is what the assignment asks for.

5. Results: loop predictor

5.1 Effect per trace

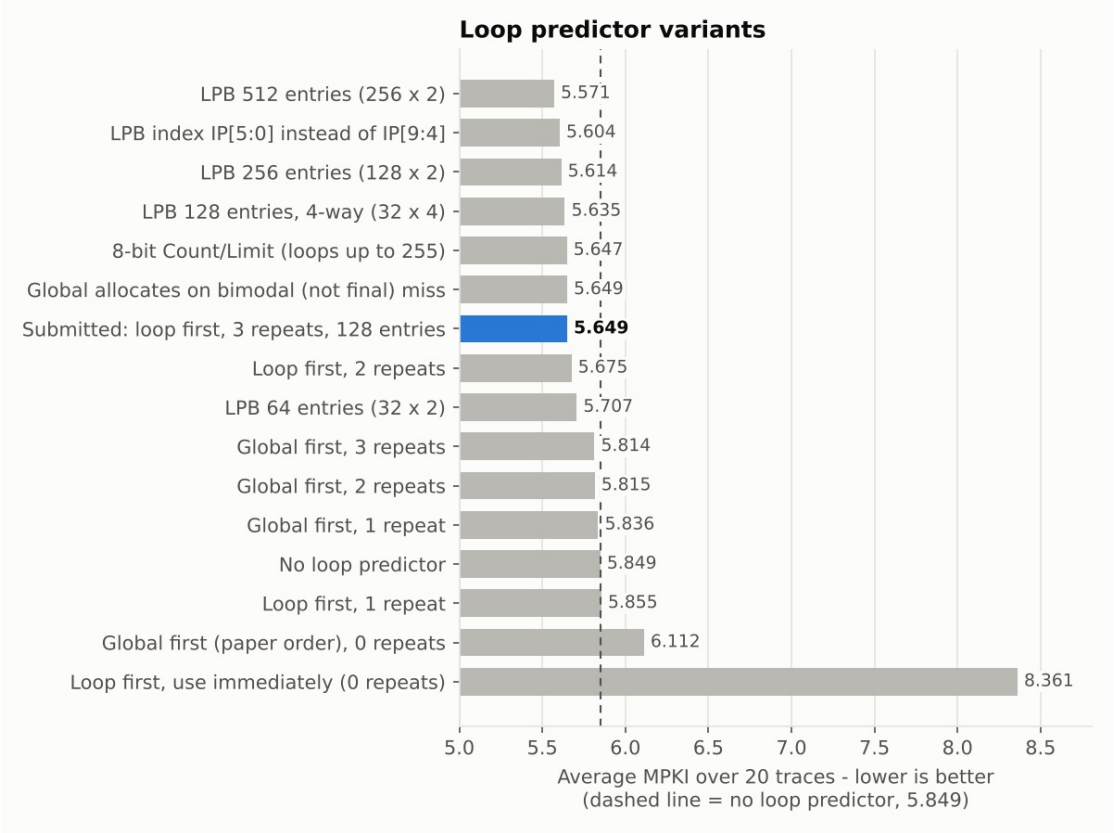


The loop predictor supplies 9.9% of all conditional predictions on average, at 98.0% accuracy. It ranges from 0.2% on bzip2 to 58.8% on eon, a C++ ray tracer full of short fixed-count loops.

Where it helps. It removes 13–20% of the remaining mispredictions on jess, javac, mpegaudio, eon and bzip2, and roughly 4–8% on gzip, compress, raytrace, db, mtrt, jack, perlbmk and twolf.

Where it hurts. It makes three traces slightly worse: mcf (+1.5%), parser (+0.9%) and vortex (+0.6%). mcf and parser have the lowest LPB accuracy of all traces (about 95%). Their loops have data-dependent trip counts that repeat a few times, which makes the entry confident, and then change, which costs one miss per change. On vortex the change is very small (+0.005 MPKI).

5.2 Design choices



- **Priority is what matters most.** With the paper's order (global hit wins), the best result is 5.814 (−0.6%), and the unit test shows why: a loop of 20 or more iterations still costs one miss per run, because the all-taken global entry predicts the exit wrong. Letting a confident LPB entry win gives 5.649 (−3.4%) and zero misses on those loops.
- **Confidence is required.** Using the LPB as soon as a Limit is learned is disastrous with loop-first priority: 8.361, worse than no loop predictor. Every non-loop branch that once mispredicted then gets "predicted" by a meaningless trip count. With global-first priority it still hurts (6.112). Requiring 2–3 repeats gives the best results; 3 is the submitted value.
- **Size.** 64 entries gives 5.707, 128 (Pentium M) 5.649, 256 5.614 and 512 5.571. Gains are modest beyond the Pentium M size.
- **Indexing.** Using IP[5:0] instead of IP[9:4] for the set index gives 5.604. The paper's index drops IP[3:0], so branches in the same 16-byte fetch block share an entry. That suits the real fetch-block-based front end, but in this per-branch simulator it causes some aliasing.
- **Little effect:** 8-bit Count/Limit, tracking loops up to 255 iterations (5.647 vs. 5.649); 4-way instead of 2-way at 128 entries (5.635); and allocating global entries on bimodal rather than final mispredictions (no measurable difference).

6. Conclusions

- A small hybrid in the style of the Pentium M (about 4 KB) beats a 32K-entry gshare (8 KB) by 10%. Tags and set-associativity remove most destructive aliasing in the global table, and the bimodal table cheaply handles the majority of branches that are simply biased.
- **Which branches the global table admits matters more than its size.** Allocating only on mispredictions is worth 25% fewer mispredictions compared with allocating on every miss.
- The **loop predictor** is a small structure (368 B) that removes 3.4% of the remaining mispredictions, and up to 20% on loop-heavy traces. It only pays off if a confident loop entry can override the global table and is used only after its trip count has repeated.

Possible next step: the BTB/iBTB target predictors (the `cpm_predictor` extra credit).

7. Code used / references

- All predictor code, both set-associative LRU tables and the tests were written from scratch for this assignment. No external cache code was used; the suggested L1-Cache-Simulator repository was not needed.
- The simulation infrastructure, traces and sample `gshare_predictor` are from the CBP-2 kit (`cbp2-infrastructure-v2`)

provided with the assignment.

- V. Uzelac and A. Milenkovic, “Experiment Flows and Microbenchmarks for Reverse Engineering of Branch Predictor Structures,” ISPASS 2009 – sizes, index/tag fields, LPB organization and the PIR update rule.
- M. Milenkovic, A. Milenkovic and J. Kulick, “Demystifying Intel Branch Predictors,” WDDD 2002 – background.
- Assignment handout, Appendix B – the hybrid predictor worked example used for verification.